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MyDesign Portfolio

Element I Initial Template

This section explains how effectively our design met the goals we set at the beginning of the project. To evaluate this, we focused on three high-priority areas: how consistently water flows through the system, how durable and portable the design is when moved, and how realistic it is for others to build and use the system with basic materials. These tests helped us figure out how well the system actually works in real-life situations, not just in theory. We collected data through multiple trials, made observations, and looked at how each part of the design performed under different conditions. Each test gave us important insights into what worked well and what still needs to be improved. The results are explained in detail below, showing what the data revealed and how close our prototype came to meeting the original design goals.

Test 1: Water Flow Consistency

- **Test Priority:** High
- **Procedure Recap:**
 - Set up the tubing and IV needle system on a PVC frame
 - Attach a temporary water source (bottles instead of an IV bag)
 - Hold the frame manually
 - Adjust IV needle pressure to control water flow
 - Observe whether water reaches the plants via the PVC holes
- **Variables:**
 - Independent Variable:** IV needle pressure
 - Dependent Variable:** Water flow rate and consistency
 - Control Variables:** Water volume, tubing length, test duration

Number of test: 3

Data: Trial 1: Water reached plants, slight leaks, inconsistent flow
Trial 2: More leaks, weak flow, water barely reached plants

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Trial 3: Improved with adjustments, water reached the plan

Results Summary: Water flow was inconsistent across trials due to setup limitations, specifically the unstable frame and the temporary use of water bottles instead of a proper IV bag. In 2 out of the 3 tests, water successfully reached the plants, but not at a steady or controlled rate. Leaks were observed in multiple areas of the tubing, causing noticeable water loss and reducing the overall efficiency of the system. These issues highlight the need for a more secure frame and a consistent, elevated water source to maintain steady pressure.

Conclusion:

Based on our testing data, we found that increased IV needle pressure helped improve water delivery, but the instability of the water source and tubing leaks reduced overall success. This means the design is not yet successful in ensuring steady water flow because water loss and inconsistency prevent reliable plant watering.

Implications:

The current frame is not practical for hands-free operation
Tubing leaks must be sealed more effectively
A steady, elevated water source (like a suspended IV bag) is critical

Next Steps:

1. Build a stable frame for the system to stand independently
2. Replace or reinforce tubing to eliminate leaks
3. Use an actual IV bag for consistent pressure and gravity-fed flow
4. Conduct 5+ trials to strengthen data validity

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Test 2: Durability and Portability

- Priority: High
- Procedure: The system was tested by carefully carrying it to five different locations and lightly shaking it by hand to simulate handling and movement. Since a frame has not yet been built, the entire setup had to be held in the air manually during testing.
- **Variables:**
Independent Variable: Movement and location changes
Dependent Variable: Stability of the structure and tubing alignment
Control Variables: Materials used, handling method, and test environment
- **Number of Trials: 5**
Moving the system across the room while shaking it gently.
Lifting and carrying the system between different positions.
Transporting the system outdoors for a short distance.
Walking with the system while shaking it lightly.
Relocating the system within the room.
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- **Results:**
Without a supporting frame, the system was completely hand-held during all tests. Despite this, the main structure—tubing and PVC piping—remained relatively stable. The PVC pipes and tubing shifted slightly with movement, but no parts detached or broke. These minor shifts can likely be fixed by sealing the joints with hot glue in the final version. Overall, while the system can't stand independently yet, it held



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together well when manually supported.

- **Conclusion of This Test:** Based on our testing, we found that the current prototype isn't portable yet because there's no frame, so it had to be held up by hand. Even though the system stayed mostly stable, it still shifted a little when moved. This shows the design is getting close, but still doesn't fully meet the goal of being durable and easy to move. Once we add a frame and seal the parts with hot glue, it should work better and meet this requirement.

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Implications: Structure lacks a stable base
Connections between parts need reinforcement
System must be redesigned to remain upright on its own during movement

- **Next Steps:**
 1. Add a weighted base or stabilizers to the
 2. Find a better source for the water and seal the edges of the PVC piping
 3. Retest under various movement scenarios with revised design

Test 3: Realistic to Build and Use

Priority: High

Procedure:

We tested if the system could be built easily using classroom materials. We asked different students to follow the instructions to build the system and timed how long it took them to adjust the IV needle pressure to get the water flow right.

Variables:

- **Independent Variable:** The students performing the setup
- **Dependent Variable:** Time to build and the functionality of the system (water flow after pressure adjustment)
- **Control Variables:** Materials used, instructions followed, and steps taken

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Number of Trials:

3 (each trial done by a different person)

Results:

- Each person was able to build the system in 8–10 minutes.
- The most challenging part was adjusting the IV needle pressure to get the water flow to the correct level.
- Once the right pressure was found, the system worked well without further adjustments.
- The materials were easy to use, and the instructions were mostly clear.

Conclusion of This Test:

Based on our testing, the design is realistic to build and use. The system was easy for students to assemble in a short amount of time. While adjusting the IV needle pressure took some time and practice, it did not require continuous changes once it was set. This indicates that the system is user-friendly, with only a small learning curve.

Implications:

- The system can be built and used without a lot of assistance, meeting the goal of simplicity.
- Adjusting the IV needle pressure should be easier with some practice, but clearer instructions could help reduce time spent on this step.

Next Steps:

1. Improve the setup instructions with step-by-step pictures to make the process clearer.
2. Mark the optimal IV needle pressure point to simplify setup.

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3. Test with a wider range of users to ensure consistency and ease of use.

